信号与系统习题

2026年4月16日 10:50

$X(s)=\int_{-\infty}^{+\infty}x(t)e^{-st}dt$

9.2 $x(t)=e^{-5t}u(t-1)$

$=e^{-5(t-1)}\cdot e^{-5}u(t-1)$

$=e^{-5}e^{-5(t-1)}u(t-1)$

$=e^{-5}e^{-s}\frac{1}{s+5}$

$=e^{-(5+s)}\cdot\frac{1}{s+5}$ ROC $s>-5$

(b) $g(t)=Ae^{5(-t-t_0)}\cdot e^{5t_0}u(-t-t_0)$

$=-Ae^{5t_0}\frac{e^{st_0}}{s+5}$

$=-A\frac{e^{(s+5)t_0}}{s+5}$ $A=-1, t_0=-1$

9.15 右边信号 $Re(s)<-5$

$\frac{dx(t)}{dt}=-2y(t)+\delta(t)$

$\frac{dy(t)}{dt}=2x(t)$

$\Rightarrow\begin{cases}sX(s)=-2Y(s)+1\\ sY(s)=X(s)\end{cases}$

$\begin{cases}Y(s)=\frac{2}{s^2+4}\\ X(s)=\frac{s}{s^2+4}\end{cases}$?

9.19. 单边拉普拉斯变换

(a) $e^{-2t}u(t+1)$ e^{-st} 缺 $=e^{-2t}u(t)$

$=\int_0^\infty e^{-2t}u(t+1)e^{-st}dt$ $=\frac{1}{s+2}$

$=\int_0^\infty e^{-(2+s)t}dt$

$=\frac{1}{2+s}(0+1)$

$=\frac{1}{2+s}$

(b) $X(s)=\int_0^\infty(\delta(t+1)+\delta(t)+e^{-2(t+3)}u(t+1))e^{-st}dt$

$=\delta(t)+e^{-2(t+3)}u(t)$

对于单边拉普拉斯变换

时其单边的特性常常"丢掉"一部分性质

9.31

(a) $s^2Y(s)-sY(s)-2Y(s)=X(s)$

$\frac{Y(s)}{X(s)}=H(s)=\frac{1}{s^2-s-2}$

$=\frac{1}{(s-2)(s+1)}$

(b) 系统稳定 双边信号 因果 右边信号 非稳 非因果

$=\left(\frac{1}{s-2}-\frac{1}{s+1}\right)\frac{1}{3}$ $-1<s<2$

$=\frac{1}{3}(-e^{2t}u(-t)-e^{-t}u(t))$

9.33 $X(s)=e^{-t}u(t)+e^{t}u(-t)$

$=\frac{1}{s+1}-\frac{1}{s-1}$

$=\frac{-2}{(s+1)(s-1)}$ ROC $1<Re\{s\}<1$

$Y(s)=X(s)H(s)$

$=\frac{-2}{(s+1)(s-1)}\frac{s+1}{(s^2+2s+2)}$

$=\frac{-2}{(s-1)(s^2+2s+2)}$

$=\frac{A}{s-1}+\frac{Bs+C}{s^2+2s+2}$

$=\frac{As^2+2As+2A+Bs^2+Cs-Bs-C}{s-1}$

$\begin{cases}A+B=0\\ 2A+C-B=0\\ 2A-C=-2\end{cases}$ $\begin{cases}A=-\frac{2}{5}\\ B=\frac{2}{5}\\ C=\frac{6}{5}\end{cases}$

$\frac{-\frac{2}{5}}{s-1}+\frac{\frac{2}{5}(s+3)}{s^2+2s+2}$

$=\frac{-\frac{2}{5}}{s-1}+\boxed{\frac{\frac{2}{5}s+1}{(s+1)^2+1}}+\frac{4}{5}\frac{}{(s+1)^2+1}$

$=\frac{2}{5}e^{t}u(-t)+\frac{2}{5}e^{-t}\cos t\,u(t)+\frac{4}{5}e^{-t}\sin t\,u(t)$

$s+1$ $\frac{s}{s^2+1}$

$\frac{s+1}{(s+1)^2+1}$

9.8 双边的 $\frac{G(j\omega)}{}$ 包含jω轴由 $e^{2t}x(t)$ 两个极点

$\Downarrow$

$X(s-2)=G(s)$

$(-1,1)$ 双边信号

9.9

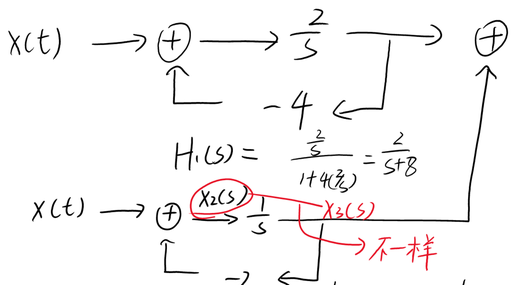
$X(s)=\frac{2(s+2)}{(s+3)(s+4)}$ 设 $\frac{A}{s+3}+\frac{B}{s+4}$

$=\frac{1}{s+3}+\frac{1}{s+4}-\frac{3}{(s+3)(s+4)}$

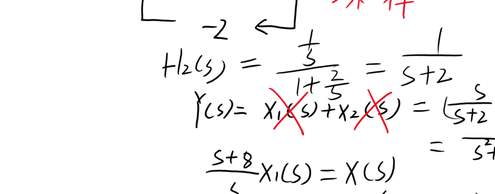
$=e^{-3t}+e^{-4t}-\left(\frac{1}{s+3}-\frac{1}{s+4}\right)\cdot 3$

$=-2e^{-3t}+4e^{-4t}$ $(e^{-3t}-e^{-4t})3$

9.17 方框图



$H_1(s)=\frac{\frac{2}{s}}{1+4(\frac{2}{s})}=\frac{2}{s+8}$



不一样

$H_2(s)=\frac{\frac{1}{s}}{1+\frac{2}{s}}=\frac{1}{s+2}$

$Y(s)=X_1(s)+X_2(s)=\left(\frac{s}{s+2}+\frac{s}{s+8}\right)X(s)$

$=\frac{s^2+10s}{s^2+10s+16}X(s)$

$\frac{s+8}{s}X_1(s)=X(s)$

$X_1(s)=\frac{s}{s+8}X(s)$

$X_2(s)=\frac{s}{s+2}X(s)$

$\therefore H(s)=H_1(s)+H_2(s)$

$=\frac{3s+12}{s^2+10s+16}$

$\therefore H(s)=\frac{Y(s)}{X(s)}$

进行拉普拉斯逆变换

9.20 (未批)

3.19 $U=L\frac{dY(t)}{dt}$ $\therefore y(t)+\frac{dy(t)}{dt}=x(t)$

(a) $I_2=\frac{U}{R}=\frac{dY(t)}{dt}$ $x(t)=e^{-2t}u(t)$

$Y(s)+sY(s)=\frac{1}{s+2}$

$Y(s)=\frac{1}{(s+2)(s+1)}$

$=\frac{1}{s-1}-\frac{1}{s+2}$

$y(t)=e^{-t}u(t)-e^{-2t}u(t)$

(b) $Y(s)+sY(s)-y(0^-)=X(s)$ 零输入 $X(s)=0$

$(s+1)Y(s)=X(s)+1$

$Y(s)=\frac{1}{s+1}X(s)+1$

(c) 当 $x(t)=e^{-2t}u(t)$

$Y(s)=2e^{-t}u(t)-e^{-2t}u(t)$

9.40 $s^3Y(s)+6s^2Y(s)+11sY(s)+6Y(s)=X(s)$

$(s^3+6s^2+11s+6)Y(s)=X(s)$

$Y(s)=\frac{}{(s^3+6s^2+11s+6)(s+4)}$

(a) 留数法 $=\frac{1}{(s^2+3s+2)(s+3)(s+4)}$

$=\frac{1}{(s+2)(s+1)(s+3)(s+4)}$

$=\frac{A}{s+1}+\frac{B}{s+2}+\frac{C}{s+3}+\frac{D}{s+4}$

$=A=\frac{1}{6}$ $B=\frac{1}{2}$ $C=-\frac{1}{2}$ $D=-\frac{1}{6}$

(b) $=-sy'(0^-)-y(0^-)$ $-6y(0^-)$

$s^3Y(s)-s^2y'(0^-)+6s^2Y(s)-6sy(0^-)+11sY(s)-11y(0^-)+6Y(s)=0$

$(s^3+6s^2+11s+6)Y(s)=s^2+5s+6$

$Y(s)=\frac{s^2+5s+6}{s^3+6s^2+11s+6}=\frac{1}{s+1}$

$=\frac{7}{6}e^{-t}u(t)+\frac{1}{2}e^{-2t}u(t)-\frac{1}{2}e^{-2t}u(t)-\frac{1}{6}e^{-4t}u(t)=e^{-t}u(t)$